5/10/2026

Assignment-3

Created By
~pintu mer

1. **Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).**



2. **Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.**

OSI Layer – Match the Following

1. **Encrypting data before sending: - Presentation Layer (Layer 6)**
2. **Routing packets across networks: - Network Layer (Layer 3)**
3. **Converting data to electrical signals: - Physical Layer (Layer 1)**
4. **Opening a web page in your browser: - Application Layer (Layer 7)**
5. **Detecting and correcting errors in frames: - Data Link Layer (Layer 2)**

3. **Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.**

Example: WhatsApp

➔ **When I send a message on WhatsApp, different OSI layers work together:**

1. **Application Layer (Layer 7): WhatsApp uses application protocols to send and receive messages.**
2. **Transport Layer (Layer 4): TCP/UDP helps transport the data between my phone and the WhatsApp server.**
3. **Network Layer (Layer 3): IP addresses help route the data from my phone to the WhatsApp server.**
4. **Data Link Layer (Layer 2): Wi-Fi/Ethernet carries the data between my device and the local network.**
5. **Physical Layer (Layer 1): Wi-Fi radio signals or cables transmit the data as electrical/radio signals.**

➔ **In short: WhatsApp → Application → Transport → Network → Data Link → Physical → Network → Server.**

4. **Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.**

Answer :

➔ **If you can ping the website's IP address but cannot open the website, the problem is likely at the Application Layer (Layer 7) or Transport Layer (Layer 4).**

1. **Network Layer (Layer 3) – Working ✅**
    - **Ping works, so IP connectivity is working.**
    - **The device can reach the website's IP address.**
2. **Transport Layer (Layer 4) – Possible Problem ⚠**
    - **The website may not be accepting TCP connections on port 80 (HTTP) or 443 (HTTPS).**
3. **Application Layer (Layer 7) – Possible Problem ⚠**
    - **The web server or HTTP/HTTPS service may be down or not responding.**

➔ **Conclusion: Since ping works, Layer 3 is working. The problem is most likely at Layer 4 or Layer 7.**